SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	**Mark**
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System

CO4-ASSESSMENT 1

Course:

Software Engineering / Code Review and Version Control Evaluation

Name:

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1. System Overview

The Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System is intended to automate and intelligently manage hotel operations while improving guest experience.

The system may include:

- Guest registration and profile management
- Room search, availability, and booking
- Check-in/check-out
- Room allocation
- Housekeeping management
- AI-based guest assistance/chatbot
- Guest requests and service management
- Food/room-service requests
- Smart-room/device control
- Complaint and feedback management
- AI-based recommendations/personalization
- Staff and task management
- Notifications
- Billing and payment
- AI analytics and operational alerts
- Role-based access and security

2. Functional Requirements to Be Tested

FR ID	Functional Requirement
FR-01	The system shall allow guests to register/login.
FR-02	The system shall validate guest credentials.

FR ID	Functional Requirement
FR-03	The system shall allow guests/staff to search room availability.
FR-04	The system shall allow authorized users to create, modify, and cancel reservations.
FR-05	The system shall prevent double booking of rooms.
FR-06	The system shall allocate an available room to a valid reservation.
FR-07	The system shall support guest check-in.
FR-08	The system shall support guest check-out and bill generation.
FR-09	The system shall calculate room/service charges correctly.
FR-10	The system shall accept valid payments and handle failed transactions.
FR-11	The system shall allow guests to request hotel services.
FR-12	The system shall assign service requests to appropriate staff.
FR-13	The system shall track service-request status.
FR-14	The system shall manage housekeeping tasks and room status.
FR-15	The AI assistant shall respond to supported guest queries.
FR-16	The AI assistant shall provide appropriate fallback responses for unsupported queries.

FR ID	Functional Requirement
FR-17	The system shall provide personalized recommendations where sufficient guest data exists.
FR-18	The system shall send appropriate notifications/reminders.
FR-19	The system shall support guest feedback and complaints.
FR-20	The system shall provide dashboards/reports for authorized staff.
FR-21	The system shall enforce role-based access control.
FR-22	The system shall maintain an audit trail for important operations.
FR-23	The system shall handle smart-room/device commands where supported.
FR-24	The system shall protect guest information and payment information.

3. Non-Functional Requirements to Be Tested

NFR ID	Requirement
NFR-01	**Performance:** Common operations should respond within an acceptable predefined time.
NFR-02	**Scalability:** The system should support increasing numbers of guests and concurrent users.
NFR-03	**Availability:** Hotel staff should be able to access the system during operational hours with minimal downtime.
NFR-04	**Security:** Unauthorized users must not access restricted functions/data.

NFR ID	Requirement
NFR-05	**Usability:** Interfaces should be understandable and easy to navigate.
NFR-06	**Reliability:** Transactions should not be lost or duplicated during normal failures.
NFR-07	**Compatibility:** The application should work across supported browsers/devices.
NFR-08	**Maintainability:** Errors should be logged clearly for administrators/developers.
NFR-09	**Accessibility:** The interface should be usable by people with different accessibility needs.
NFR-10	**AI Quality:** AI responses should be relevant, understandable, and appropriately cautious when uncertain.
NFR-11	**Data Integrity:** Reservation, room, guest, service, and billing data should remain consistent.
NFR-12	**Privacy:** Guest information should only be displayed to authorized users.

4. Major Test Scenarios

Guest Management

- Guest registration
- Login/logout
- Invalid credentials
- Password validation
- Profile modification

Reservation Management

- Room search
- Available room selection
- Booking creation

- Booking modification
- Booking cancellation
- Double-booking prevention
- Invalid dates
- Maximum occupancy

Check-in/Check-out

- Valid check-in
- Early/late check-in
- Invalid reservation
- Check-out
- Outstanding payment
- Bill generation

Housekeeping

- Room status update
- Housekeeping task creation
- Task assignment
- Task completion
- Room unavailable during maintenance

Guest Services

- Service request
- Request prioritization
- Staff assignment
- Request cancellation
- Request status tracking

AI Assistant

- FAQ response
- Reservation-related query
- Service-request query

- Personalized recommendation
- Ambiguous query
- Unsupported question
- AI failure/fallback

Payment

- Successful payment
- Failed payment
- Invalid payment data
- Duplicate payment prevention
- Refund/cancellation

Smart Room

- Device control
- Invalid command
- Device unavailable
- Multiple simultaneous commands

Security

- Role-based access
- Session timeout
- Unauthorized API/function access
- Data privacy
- Audit logging

5. Detailed Test Cases

Note: Since these are test-design cases prepared before execution, Actual Result and Pass/Fail Status should initially be recorded as Not Executed. They should be updated during actual testing.

A. Guest Registration and Authentication

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-001	Register with valid details	Open registration → enter details → submit	Name: Arun; valid email/mobile; valid password	Account created successfully	Not Executed	Not Executed
TC-002	Register with mandatory field empty	Leave name/email empty → submit	Name = blank	Appropriate validation message displayed	Not Executed	Not Executed
TC-003	Register using existing email	Enter already registered email → submit	Existing email	Registration rejected with suitable message	Not Executed	Not Executed
TC-004	Invalid email format	Enter invalid email → submit	arun@	Email validation message displayed	Not Executed	Not Executed
TC-005	Password below minimum length	Enter short password → submit	Password shorter than configured minimum	Password rejected	Not Executed	Not Executed
TC-006	Password at minimum boundary	Enter password with exactly minimum allowed length	Exactly minimum length	Password accepted if other rules are satisfied	Not Executed	Not Executed
TC-007	Login with valid credentials	Enter valid username/password → login	Valid account	User successfully logged in	Not Executed	Not Executed
TC-008	Login with incorrect password	Enter valid username + incorrect password	Correct username, wrong password	Login rejected	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-009	Login with nonexistent account	Enter unregistered credentials	Unknown account	Login rejected without exposing sensitive account information	Not Executed	Not Executed
TC-010	Logout	Login → select Logout	Valid session	Session terminated and login screen displayed	Not Executed	Not Executed

Techniques: Equivalence Partitioning, Boundary Value Analysis.

B. Room Search and Reservation

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-011	Search available room	Enter valid dates and occupancy → search	10 Sep–12 Sep, 2 guests	Available rooms displayed	Not Executed	Not Executed
TC-012	Search when no room is available	Enter fully booked dates	Date with no availability	System displays no-availability message and alternatives if supported	Not Executed	Not Executed
TC-013	Check-in date before current date	Enter past check-in date	Past date	Search rejected with validation message	Not Executed	Not Executed
TC-014	Check-out before check-in	Enter invalid date order	Check-in: 15 Sep;	Reservation/search rejected	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
			checkout: 12 Sep			
TC-015	Same-day valid reservation	Enter valid same-day dates if hotel policy allows	Same check-in/check-out date	System follows configured hotel policy	Not Executed	Not Executed
TC-016	Occupancy within room capacity	Search using supported occupancy	2 guests for double room	Room displayed as available	Not Executed	Not Executed
TC-017	Occupancy exceeds capacity	Enter occupants above room capacity	5 guests for 2-person room	Room rejected/not shown	Not Executed	Not Executed
TC-018	Create valid reservation	Select available room → enter guest information → confirm	Available room, valid guest	Reservation created with unique ID	Not Executed	Not Executed
TC-019	Attempt double booking	Two users attempt to reserve same final available room simultaneously	Same room/date	Only one reservation succeeds; other receives updated availability	Not Executed	Not Executed
TC-020	Cancel valid reservation	Open reservation → Cancel → confirm	Existing cancellable booking	Reservation cancelled according to policy	Not Executed	Not Executed
TC-021	Modify reservation	Open booking → change dates → save	New available dates	Reservation updated successfully	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-022	Modify to unavailable dates	Change booking to fully booked dates	Unavailable dates	Modification rejected and existing booking remains unchanged	Not Executed	Not Executed

Techniques: Equivalence Partitioning, Boundary Value Analysis, Decision Table, Concurrency testing.

C. Check-in and Check-out

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-023	Valid guest check-in	Search reservation → verify guest → check in	Valid confirmed reservation	Guest checked in and room status changes appropriately	Not Executed	Not Executed
TC-024	Check-in without reservation	Attempt walk-in/check-in without valid booking	Unknown reservation	System follows configured walk-in policy or rejects operation	Not Executed	Not Executed
TC-025	Check-in cancelled reservation	Enter cancelled reservation	Cancelled booking	Check-in rejected	Not Executed	Not Executed
TC-026	Check-in unavailable room	Attempt allocation to maintenance/occupied room	Room status = unavailable	System prevents allocation	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-027	Valid check-out	Select checked-in guest → checkout	Active stay	Checkout completed and bill generated	Not Executed	Not Executed
TC-028	Check-out with outstanding balance	Checkout guest with unpaid charges	Outstanding balance	System follows payment policy and prevents completion if required	Not Executed	Not Executed
TC-029	Generate bill with room + service charges	Add charges → checkout	Room ₹5,000 + services ₹1,000	Total calculated correctly according to configured taxes/rules	Not Executed	Not Executed
TC-030	Check-out already checked-out guest	Attempt second checkout	Completed stay	Operation rejected safely	Not Executed	Not Executed

D. Payment and Billing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-031	Successful payment	Select payment method → submit valid payment	Valid test payment	Payment succeeds and transaction recorded	Not Executed	Not Executed
TC-032	Failed payment	Submit payment that gateway rejects	Declined test transaction	Failure message displayed; booking/bill	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
				remains consistent		
TC-033	Invalid payment details	Enter invalid test payment data	Invalid card/test payment information	Validation/error message displayed	Not Executed	Not Executed
TC-034	Duplicate payment submission	Click Pay twice rapidly	Same transaction	System prevents duplicate charge	Not Executed	Not Executed
TC-035	Payment timeout	Interrupt/timeout payment gateway response	Simulated timeout	System displays appropriate status and does not create duplicate transaction	Not Executed	Not Executed
TC-036	Refund	Cancel eligible paid booking → initiate refund	Valid refundable transaction	Refund processed according to policy and transaction status updated	Not Executed	Not Executed

E. Housekeeping and Hotel Operations

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-037	Create housekeeping task	Mark room for cleaning	Room 201	Task created	Not Executed	Not Executed
TC-038	Assign task to staff	Select task → assign staff	Available housekeeping employee	Task assigned successfully	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-039	Complete housekeeping task	Staff opens task → marks completed	Room 201	Task completed and room status updated	Not Executed	Not Executed
TC-040	Assign task to unavailable staff	Select staff marked unavailable	Staff status = unavailable	Assignment rejected or rerouted	Not Executed	Not Executed
TC-041	Maintenance room cannot be booked	Mark room under maintenance → search	Room 305 = maintenance	Room excluded from available inventory	Not Executed	Not Executed
TC-042	Concurrent room-status updates	Two staff update same room	Conflicting status changes	System preserves data consistency and handles conflict appropriately	Not Executed	Not Executed

F. Guest Service Requests

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-043	Submit valid service request	Guest selects service → submit	Extra towel	Request created with tracking ID	Not Executed	Not Executed
TC-044	Submit request without required information	Leave required field empty → submit	Missing room/request details	Validation message displayed	Not Executed	Not Executed
TC-045	Assign service request	Staff views pending	Valid staff member	Request assigned	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
		requests → assign				
TC-046	Update request status	Staff changes status	Pending → In Progress → Completed	Status changes correctly	Not Executed	Not Executed
TC-047	Cancel request	Guest cancels eligible request	Existing pending request	Request cancelled according to rules	Not Executed	Not Executed
TC-048	High-priority request	Submit configured urgent request	Medical/emergency-type request*	System follows configured escalation process rather than treating it as an ordinary request	Not Executed	Not Executed

*The actual application should define appropriate emergency handling and escalation rules.

G. AI Guest Assistant

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-049	Ask supported hotel FAQ	Open AI assistant → ask question	"What time is breakfast?"	Relevant answer returned based on current hotel information	Not Executed	Not Executed
TC-050	Ask about room availability	Ask AI about rooms	"Are rooms available	AI provides availability or directs user to	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
			for my dates?"	booking flow using current data		
TC-051	Create service request through AI	Ask AI for a supported service	"I need extra towels"	AI confirms/creates request if integration is supported	Not Executed	Not Executed
TC-052	Ambiguous query	Enter unclear question	"Can you arrange it?"	AI asks an appropriate clarification question	Not Executed	Not Executed
TC-053	Unsupported question	Ask unrelated/unsupported question	Unrelated query	AI politely indicates limitation or redirects appropriately	Not Executed	Not Executed
TC-054	AI service unavailable	Disable AI service → submit query	Normal guest query	System shows fallback/error message without crashing	Not Executed	Not Executed
TC-055	Outdated hotel information	Modify hotel information → ask AI	Current hotel policy	AI should use the current approved information rather than stale information	Not Executed	Not Executed
TC-056	Multiple consecutive queries	Submit multiple valid questions	10 sequential questions	AI remains responsive and maintains appropriate	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
				conversation context		

H. AI Personalization and Recommendations

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-057	Recommendation with sufficient guest history	Load guest profile → request recommendation	Previous dining/service preferences	Relevant recommendations generated	Not Executed	Not Executed
TC-058	Recommendation with no history	New guest requests recommendation	Empty preference history	Generic/relevant recommendations provided without inventing preferences	Not Executed	Not Executed
TC-059	Recommendation using invalid data	Insert malformed profile preference data	Invalid preference record	System handles data safely and does not crash	Not Executed	Not Executed
TC-060	AI recommendation unavailable	Disable recommendation service	Valid guest	System provides fallback recommendations or standard options	Not Executed	Not Executed

I. Smart-Room Features

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-061	Turn supported device on	Guest selects device → ON	Room light/device	Device changes to ON	Not Executed	Not Executed
TC-062	Turn device off	Select device → OFF	Room light/device	Device changes to OFF	Not Executed	Not Executed
TC-063	Device unavailable	Send command while device is offline	Offline device	User receives suitable error/status message	Not Executed	Not Executed
TC-064	Unauthorized device control	User outside authorized room attempts control	Wrong room/session	Command rejected	Not Executed	Not Executed
TC-065	Invalid device command	Send unsupported command	Unsupported command	Command rejected safely	Not Executed	Not Executed

J. Feedback and Complaints

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-066	Submit valid feedback	Select rating → enter feedback → submit	Rating 5, valid text	Feedback recorded	Not Executed	Not Executed
TC-067	Rating below minimum	Submit rating below configured range	Rating 0 if range is 1–5	Validation error	Not Executed	Not Executed
TC-068	Rating at boundaries	Submit minimum and	1 and 5	Both accepted	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
		maximum valid ratings				
TC-069	Submit empty feedback	Submit without required information	Blank feedback	Appropriate validation displayed	Not Executed	Not Executed
TC-070	Complaint tracking	Submit complaint → view status	Valid complaint	Complaint receives ID and can be tracked	Not Executed	Not Executed

K. Security and Access Control

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-071	Guest accesses guest functions	Login as guest → open guest features	Guest account	Authorized functions accessible	Not Executed	Not Executed
TC-072	Guest attempts admin function	Login as guest → open admin dashboard	Guest account	Access denied	Not Executed	Not Executed
TC-073	Staff accesses permitted functions	Login as staff	Staff account	Authorized functions accessible	Not Executed	Not Executed
TC-074	Unauthenticated access	Directly open protected page	No active session	User redirected/denied	Not Executed	Not Executed
TC-075	Session timeout	Login → remain inactive beyond	Expired session	Session expires and protected data is inaccessible	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
		configured timeout				
TC-076	Audit log generation	Perform booking modification	Valid staff account	Important operation recorded in audit log	Not Executed	Not Executed
TC-077	Guest privacy	Guest A attempts to view Guest B's details	Different guest accounts	Access denied	Not Executed	Not Executed

6. Decision Table – Room Reservation

A decision table can be used to test reservation eligibility.

Condition/Action	Rule 1	Rule 2	Rule 3	Rule 4
Valid guest	Y	Y	N	Y
Valid dates	Y	N	Y	Y
Room available	Y	Y	Y	N
Occupancy valid	Y	Y	Y	Y
Create reservation	Yes	No	No	No
Display validation/error	No	Yes	Yes	Yes

Example Test Cases

- **Rule 1:** All conditions valid → booking should succeed.
- **Rule 2:** Invalid dates → booking should be rejected.
- **Rule 3:** Invalid guest → booking should be rejected.
- **Rule 4:** Room unavailable → booking should be rejected.

7. State Transition Testing

Room status can be modeled as:

Available → Reserved → Occupied → Cleaning → Available

Alternative exceptional states:

Available → Maintenance → Available

State Transition Test Cases

TC ID	Current State	Event	Expected Next State
ST-01	Available	Valid reservation	Reserved
ST-02	Reserved	Guest checks in	Occupied
ST-03	Occupied	Guest checks out	Cleaning
ST-04	Cleaning	Housekeeping completed	Available
ST-05	Available	Maintenance started	Maintenance
ST-06	Maintenance	Maintenance completed	Available
ST-07	Occupied	Attempt new reservation	Rejected
ST-08	Maintenance	Attempt guest check-in	Rejected

This ensures invalid state transitions are also tested.

8. Boundary Value Analysis

Boundary testing is particularly useful for:

Guest count

If a room supports 1–2 guests:

Input	Expected
0	Invalid
1	Valid
2	Valid
3	Invalid

Rating

If rating range is **1–5**:

Input	Expected
0	Invalid
1	Valid
5	Valid
6	Invalid

Password

If the configured password length is **8–20 characters**:

Length	Expected
7	Invalid
8	Valid
20	Valid
21	Invalid

Reservation duration

If hotel policy permits a maximum configured stay of **30 days**:

Duration	Expected
0 days	Invalid
1 day	Valid
30 days	Valid
31 days	Invalid

The actual boundary values should be replaced with the hotel's requirements if they differ.

9. Equivalence Partitioning

For room occupancy, possible partitions include:

Partition	Example	Expected
Valid	1–2 guests	Accepted
Below minimum	0 guests	Rejected
Above maximum	3+ guests	Rejected
Non-numeric	"two"	Rejected
Negative	-1	Rejected

For reservation dates:

Partition	Example	Expected
Valid future dates	10–12 Sep	Accepted
Past date	1 Jan	Rejected
Checkout before check-in	15 Sep → 12 Sep	Rejected
Same day	10 Sep → 10 Sep	Depends on hotel policy
Invalid format	ABC	Rejected

10. Non-Functional Test Cases

Performance

TC ID	Test Scenario	Test Steps	Expected Result
NFR-T01	Room search response time	Perform room search under normal load	Response meets specified performance target
NFR-T02	Concurrent booking requests	Simulate multiple users searching/booking simultaneously	System remains responsive and prevents double booking
NFR-T03	AI response performance	Send normal guest queries	Response meets defined AI response-time target
NFR-T04	Dashboard loading	Open staff dashboard with large dataset	Dashboard loads within specified target

Reliability

TC ID	Scenario	Expected Result
NFR-T05	Database connection interruption during booking	No incomplete/duplicate reservation
NFR-T06	Payment service interruption	Transaction state remains consistent
NFR-T07	AI service interruption	Application continues functioning with fallback

Security

TC ID	Scenario	Expected Result
NFR-T08	Unauthorized API access	Request denied
NFR-T09	Expired session access	Access denied
NFR-T10	Access another guest's information	Request denied
NFR-T11	Sensitive information exposure	Sensitive information is appropriately protected

Usability

TC ID	Scenario	Expected Result
NFR-T12	First-time guest booking	User can complete booking without unnecessary confusion
NFR-T13	Error message usability	Error messages clearly explain what needs correction
NFR-T14	Mobile interface	Core functions remain usable on supported mobile devices
NFR-T15	Accessibility	Supported accessibility features function correctly

11. Exceptional and Failure Conditions

The following conditions should specifically be included in system testing:

1. Database unavailable.
2. Network connection interrupted.
3. Payment gateway unavailable.
4. AI service unavailable.
5. Smart-room device offline.
6. Two users attempting to book the same room simultaneously.
7. Staff member becoming unavailable after task assignment.
8. Session expiration during booking.
9. Browser/app closed during payment.
10. Server restart during a transaction.
11. Invalid or corrupted input data.
12. Notification service unavailable.
13. Partial failure while updating room status.
14. Duplicate requests caused by repeated button clicks.

The expected behavior should be graceful failure, data consistency, meaningful error handling, and recovery where applicable.

12. Test Coverage Matrix

Requirement	Covered By
FR-01 Registration	TC-001 to TC-006
FR-02 Authentication	TC-007 to TC-010
FR-03 Room search	TC-011 to TC-017
FR-04 Reservation management	TC-018, TC-020–TC-022
FR-05 Double-booking prevention	TC-019, NFR-T02
FR-06 Room allocation	TC-023–TC-026

Requirement	Covered By
FR-07 Check-in	TC-023–TC-026
FR-08 Check-out	TC-027–TC-030
FR-09 Billing	TC-029
FR-10 Payment	TC-031–TC-036
FR-11 Service requests	TC-043–TC-048
FR-12 Staff assignment	TC-045
FR-13 Request tracking	TC-046–TC-047
FR-14 Housekeeping	TC-037–TC-042
FR-15 AI assistant	TC-049–TC-056
FR-16 AI fallback	TC-053–TC-054
FR-17 Personalization	TC-057–TC-060
FR-18 Notifications	Scenario coverage; dedicated notification tests recommended
FR-19 Feedback/complaints	TC-066–TC-070
FR-20 Reports	NFR-T04
FR-21 Access control	TC-071–TC-077
FR-22 Audit trail	TC-076
FR-23 Smart room	TC-061–TC-065
FR-24 Data protection	TC-077, NFR-T11

13. Test Coverage Summary

The proposed test suite provides coverage across:

Testing Area	Coverage
Functional requirements	Registration, reservation, rooms, check-in/out, billing, services, housekeeping, AI, smart room, feedback, security
Positive testing	Valid inputs and successful workflows
Negative testing	Invalid inputs and unauthorized operations
Boundary testing	Password, occupancy, ratings, stay duration
Equivalence partitioning	Dates, occupancy, credentials, ratings
Decision table testing	Reservation eligibility
State transition testing	Room lifecycle
Exception testing	Service/database/payment/AI failures
Security testing	Authentication, authorization, privacy
Performance testing	Search, AI, dashboard, concurrency
Usability testing	Guest workflows and error messages
Reliability testing	Service/database failures
Data integrity	Reservations, payments, room states

14. Entry and Exit Criteria

Entry Criteria

Testing can begin when:

- Requirements are approved.
- Test environment is available.
- Application build is deployed.

- Test database is prepared.
- Required test accounts are available.
- External integrations have suitable test/sandbox environments.
- Test data has been prepared.

Exit Criteria

Testing may be considered complete when:

- All planned test cases have been executed.
- Critical/high-severity defects are resolved or formally accepted.
- Regression testing is completed.
- Functional requirements have adequate coverage.
- Major security and data-integrity tests pass.
- Performance results satisfy the defined requirements.
- Test results and defect reports are documented.

15. Defect Severity Classification

Severity	Description	Example
Critical	System/business operation is severely blocked	Confirmed payment but no reservation and no recoverable transaction state
High	Major feature is unusable	Guest cannot complete any reservation
Medium	Important feature has a workaround	Service request status does not refresh automatically
Low	Minor issue	Small UI alignment problem

16. Recommended Test Execution Format

During actual execution, the final test-case table should use:

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-001	Valid registration	Enter valid details and submit	Valid guest data	Account created	*Enter observed result*	*Pass/Fail*
TC-002	Invalid email	Enter invalid email and submit	abc@	Validation displayed	*Enter observed result*	*Pass/Fail*
TC-003	Valid booking	Search → select room → confirm	Valid future dates	Booking created	*Enter observed result*	*Pass/Fail*

Important: Do not pre-fill "Actual Result" as the expected result. It should contain what the application actually did during execution.

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17. Conclusion

The proposed test design provides a structured approach for validating the **Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System**. It covers the complete hotel workflow—from guest registration and reservation through check-in, services, housekeeping, smart-room operations, payment, checkout, feedback, and AI-assisted guest interaction.

The test suite combines **positive, negative, boundary, exceptional, security, performance, usability, decision-table, equivalence-partitioning, and state-transition testing**. This provides broad functional and non-functional coverage and helps ensure that the system is **correct, reliable, secure, usable, and capable of maintaining consistent hotel and guest data under normal and exceptional conditions**.